

Autoplay on Trial: An Online Experiment on the Suspect Behind Excessive Screen Time

Reha Tuncer*

July 25, 2025

Abstract

Interface design features can ‘nudge’ consumers to take certain actions and are often accused of promoting addictive online behavior. A prevalent design feature across popular social media and streaming platforms is the default autoplay feature. We present an incentivized online experiment to investigate whether the autoplay feature alone can cause an increase in undesired video consumption, and elicit the willingness to pay for commitment against autoplay. In a two-day study, we recruited 236 participants to allocate 20 minutes between a transcription task and watching funny animal videos. Time allocation decisions were made hypothetically a day before and realized the next day. We randomly assigned participants to either autoplay or click-to-play media controls while keeping the video content constant. We find that the autoplay feature, in isolation, does not override participants’ previously indicated preferences for media consumption. In addition, participants exhibit a positive willingness-to-pay for autoplay (6.72 pence/hour), viewing it

*I am grateful to Ernesto Reuben and Kerstin Bongard-Blanchy for their guidance in the design and conception of this experiment, to Sophie Doublet for her help in interface design, to Suvadeep Mukherjee for his guidance in building web experiments. I am also thankful for the invaluable comments from the Behavioral Economics Summer School at the University of Copenhagen, the doctoral workshop at UC Louvain, the Experimentetrics Workshop Soletto at Sapienza University Rome, Samuel Greiff, Manuel Munoz, Hande Erkut and countless colleagues.

as a convenience feature rather than a self-control problem in our setting. We also find that experimenter demand effects and lack of content appeal result in participants allocating more time to the transcription task than planned, confounding the effect of the intended autoplay manipulation. While these findings highlight the fundamental challenge that interface features cannot be studied in isolation from their embedded context, the effectiveness of autoplay seem to depend on the personalized content it delivers.

1 Introduction

Autoplay is a user interface feature that automatically transitions users to new video content without any action. It has become a familiar design element in major social media and streaming platforms over the last decade,¹ and has been described as a default nudge that reduces the autonomy of users (Lukoff et al., 2021; Schaffner et al., 2025). Autoplay’s widespread adoption and concerns about its restriction of users’ freedom of choice have also attracted regulatory scrutiny, with the U.S. SMART Act proposing to ban platforms from serving autoplay content “without an express, separate prompt by the user” due to suspicions of its contribution to addictive online behavior.²

While autoplay is pervasive and tends to increase viewing times (Chen et al., 2025; Hiniker et al., 2018; Schaffner et al., 2025), its impact on user welfare and the mechanisms driving this effect are not well understood. Does the increase in consumption go against users’ wishes, or does it merely remove frictions to help them achieve their desired consumption levels? Can we attribute the increase in consumption to autoplay in isolation, or does it also depend on the supply of content? One reason for these gaps is the challenge of holding constant the video content displayed to users while eliciting their preferences related to content consumption before the actual consumption occurs.

We overcome this challenge in an incentivized two-day online experiment designed

¹As of 2025 these platforms include Facebook, Instagram, Netflix, YouTube, TikTok, and Twitter (currently X). Twitter first introduced autoplay in 2015, and described it as a means to reduce “extra effort” in the number of clicks/taps required to consume content.

²SMART Act of 2019, S. 2314, 116th Congress.

to isolate the causal effect of autoplay on content consumption. We recruited 184 participants who twice allocated 1200 seconds (20 minutes) between two tasks. The tasks involved seconds of transcription of random character sequences at 0.15 pence/second, and seconds of watching funny animal videos at 0.1 pence/second. On the first day, participants got familiar with tasks and made hypothetical allocations using the strategy method. On the following day, participants realized their time allocation decision without constraints. This two-day structure with hypothetical and realized choices formed the foundation for testing our preregistered hypotheses regarding autoplay.

To isolate the causal effect of autoplay on these allocation decisions, we randomly assigned participants to either the *Autoplay* or *Control* condition. In the *Autoplay* condition watching task videos played continuously. In the *Control* condition participants explicitly clicked to play each video. We held constant the content by presenting 80 manually curated “funny animal” videos from YouTube and TikTok in identical sequence. This eliminated algorithmic content curation and personalization as potential confounds. We dedicated the first block of the experiment to measure the differences between planned and realized allocations across conditions.

In a second block of the experiment, we modified the design to elicit demand for a commitment device. We recruited 52 additional participants who made nine binary choices: they could either keep their preferred autoplay setting (on or off) or switch to the opposite setting in exchange for bonus payments ranging from £0.05 to £0.50. We implemented one of these decisions at random, determining both the autoplay setting for the next day and the associated bonus payment. This allowed us to measure participants’ Willingness To Pay (WTP) for autoplay in advance, providing insight into whether participants viewed the feature as beneficial (positive WTP) or as a self-control problem requiring a commitment device (negative WTP).

We report three key findings. The first set of results focuses on our preregistered hypotheses regarding autoplay. We find no evidence that autoplay in isolation increased content consumption. Participants in the *Autoplay* condition watched statistically indistinguishable numbers of videos compared to the *Control* condition. *Autoplay* condition

also had no significant effect on the proportion of time spent on either task. These results also hold for other engagement metrics (i.e., average session length, number of sessions).

Second, participants viewed autoplay as beneficial rather than problematic. Contrary to our hypothesis that participants would demand commitment devices against autoplay, we find a positive WTP for the feature. Using data from the second block, we are able to estimate an average WTP of 6.72 pence/hour. Autoplay is considered as a convenience feature rather than as a threat to self-control, aligning with earlier research (Bongard-Blanchy et al., 2021; Schaffner et al., 2025).

Third, participants consumed less content than initially planned. Comparing hypothetical time allocations to realized ones, we find that participants consumed significantly less content than their day 1 plans implied in both conditions. The decrease amounts to 15.8 percentage points (35%) less time spent on the watching task. This result is driven by 33% of all participants who had planned to spend up to three quarters of their time watching videos but instead decreased their content consumption on the next day. On the other end, 88% of those who had initially planned to spend less than a quarter of their time watching videos successfully followed their plans and did not succumb to temptation.

Evidence from a regression discontinuity analysis shows that participants monitored their day 1 plans despite ultimately exceeding them. When participants reached their planned transcribing duration, they immediately reduced transcribing probability by 25.5 percentage points in the next 60-second window. However, this effect dissipated quickly and became statistically insignificant within 90 seconds as participants resumed transcribing.

Taken together, these findings cast doubt on whether the watching task was actually perceived as tempting. A mixed-methods analysis of open-ended responses reveals that participants perceived the experimental environment as a work setting rather than a choice between independent tasks. While 29% spontaneously described transcribing using negative terms, another 23% characterized video-watching as “taking breaks” from

work. This suggests that experimenter demand effects may have interfered with the intended effects of the autoplay manipulation, with implications for both the interpretation of our null results and the broader generalizability of our study.

A primary internal validity concern is that experimenter demand effects favored allocating more time to the transcribing task than planned. These can emerge through contextual cues even without explicit behavioral instructions, as participants interpret the unfamiliar experimental environment by drawing inferences from the information provided. They may alter behavior simply by having attention drawn to variables of interest (Zizzo, 2010). In our experiment, these could be (i) the small earnings difference favoring the transcribing task,³ (ii) multiple mentions of the quality criteria for the transcribing task to receive earnings, (iii) placement of the transcribing task as the first activity upon starting a session. Alternative designs could have addressed these concerns by rewarding balanced time allocations more, or by eliminating earning differences altogether.

Another major challenge in isolating the effects of autoplay is the role of content appeal. Higher content appeal would make the watching task more tempting, and create a stronger tradeoff between the two tasks. To address this while keeping the content identical across conditions, we conducted pretests from the same participant pool where the funny animal videos came out as the most popular theme. Participants also rated the curated videos positively (median 7/10) in a post experiment survey. Regardless, the watching task content may not have been sufficiently appealing. Allowing participants to select among a list of themes with manually curated videos could have addressed this issue (Ek & Samahita, 2023).

A final concern in online experiments is ensuring genuine participation equivalent to laboratory conditions. We implemented several measures to address this challenge. We used JavaScript-based attention monitoring to detect each second participants navigated away from the experimental interface following (Purohit et al., 2023). We also required

³This difference is 60 pence over 20 minutes, equivalent to a 1.8 pound hourly wage difference (15% above our 12 pound minimum wage baseline).

127 minimum internet speeds and restricted participation to desktop users with Chromium- 127
128 based browsers to standardize the user experience. Participants also answered post- 128
129 experiment survey questions about their engagement while watching videos and about 129
130 connection interruptions. These controls ensured that our final dataset is composed of 130
131 participants who were active, and approximate the controlled conditions of laboratory 131
132 experiments ([Arechar et al., 2018](#)). 132

133 More broadly, our findings highlight a fundamental challenge in studying interface 133
134 design features from social media and streaming platforms in controlled laboratory set- 134
135 tings. Interface elements like autoplay may be inherently difficult to evaluate in isolation: 135
136 Recent evidence suggests that the power of autoplay may lie in its coupling with recom- 136
137 mendation algorithms that serve compelling material tailored to individual preferences 137
138 ([Hiniker et al., 2018](#); [Schaffner et al., 2025](#)). Laboratory experiments that hold content 138
139 constant, as ours necessarily did for experimental control, risk ultimately capturing only 139
140 part of how these design elements operate in their natural environment. 140

141 We contribute to several strands of literature. First, the literature on studying the 141
142 effects of social media and streaming services on user well-being ([Allcott et al., 2020](#), 142
143 [2022](#); [Bao, 2025](#); [Beknazar-Yuzbashev et al., 2025](#); [Braghieri et al., 2022](#); [Bursztyn et](#) 143
144 [al., 2023](#); [Collis & Eggers, 2022](#); [Groshek et al., 2018](#); [Nyhan et al., 2023](#); [Purohit et al.,](#) 144
145 [2023](#); [Walton-Pattison et al., 2018](#)). We contribute to this work by shedding light on 145
146 one potential driver of the harmful effects of social media and streaming services, the 146
147 autoplay, in a controlled setting. 147

148 A subset of this literature focuses on the effects of specific interface design elements 148
149 on user behavior, particularly related to user agency and overconsumption of content 149
150 ([Bongard-Blanchy et al., 2021](#); [Chen et al., 2025](#); [Hoong, 2021](#); [Lukoff et al., 2021](#); 150
151 [Lupiáñez-Villanueva et al., 2022](#); [Lyngs et al., 2019, 2020](#); [Mathur et al., 2021](#); [Schaffner](#) 151
152 [et al., 2023, 2025](#); [Silverman et al., 2024](#)). To our knowledge, our paper provides the 152
153 first attempt at isolating the causal effects of autoplay on content consumption. 153

154 Methodologically, this paper contributes to the literature on laboratory experiments 154
155 that observe behavior in real time while participants trade cognitive effort with leisure 155

(Bhatia et al., 2021; Bonein & Denant-Boèmont, 2015; Ek & Samahita, 2023; Houser et al., 2018; Kool & Botvinick, 2014). We also add to a burgeoning literature manipulating design elements within the user interface to reduce overconsumption of content (Hiniker et al., 2018; Lyngs et al., 2020; Purohit et al., 2023; Schaffner et al., 2025).

The remainder of this paper is organized as follows. Section 2 describes the experimental design. Section 3 details the experimental sample and procedures. Section 4 presents our results and Section 5 concludes.

2 Design

2.1 Timeline and Session Structure

We conducted a two-day online experiment to examine how autoplay feature affects time allocation between transcription of random character sequences and watching funny animal videos. The multi-day structure allowed us to measure the gap between stated preferences and actual behavior under two different video interface conditions. Participants completed two separate data collection sessions with a cooling-off period of 24 hours (up to 48 hours) between sessions.

Day 1 sessions included a practice session to familiarize participants with the experimental interface, followed by preference elicitation measures. Day 2 featured the main 20-minute session where participants made actual time allocation decisions. We conducted two waves of data collection during May 2023, as described in Table 1.

Table 1: Summary of experimental design and timeline

	Practice Session	Time Choice	Main Session	Multiple Price List	Decision Horizon
Day 1 (May 10)	✓	✓			24-hour
Day 2 (May 11)			✓		immediate
Day 1 (May 18)	✓	✓		✓	24-hour
Day 2 (May 19)			✓		immediate

Note: This table shows the experimental components administered across two data collection waves, with columns ordered chronologically within each day. The first block (May 10-11 2023) established baseline effects, while the second block (May 18-19 2023) incorporated the Multiple Price List to measure demand for a commitment device. Decision horizon refers to the time between hypothetical preference elicitation and the main experimental session.

2.2 Technical Implementation and Interface Design

We designed the experimental platform to provide precise control over the testing environment while maintaining ecological validity. We presented both tasks within a single web page using distinct tabs, allowing participants to switch between activities seamlessly while preserving progress. Sessions began with the transcription task as the default, requiring an active choice to switch to the watching task.

The interface incorporated several key features to track behavior. A progress bar and timer provided real-time feedback on session duration. We logged all critical interactions including tab switches, keystrokes during transcription, video controls, and cursor movements outside the experimental window. This granular tracking enabled us to construct detailed measures of attention and engagement across tasks.

2.3 Internal Validity Controls

Online experiments present unique challenges for maintaining experimental control, including participant multitasking, technical disruptions, and environmental variation. We

189 implemented comprehensive measures to address these concerns and ensure data quality. 189

190 **Participant and Technology Screening:** We selectively admitted participants 190
191 using multiple screening criteria to ensure stable experimental conditions. Internet speed 191
192 requirements filtered out participants with unstable connections (minimum 30 Mbps on 192
193 day 1, 25 Mbps on day 2). We restricted participation to users with Windows, Linux, 193
194 or Mac operating systems using Chromium-based browsers exclusively. We excluded 194
195 Safari and Mozilla Firefox due to restrictive autoplay configurations. Mobile devices 195
196 and tablets were prohibited to standardize the user experience. 196

197 **Platform Controls:** The experimental website enforced linear navigation, prevent- 197
198 ing participants from moving backward through the experiment. We disabled right-click 198
199 functions and keyboard shortcuts. Any attempt to navigate backward reset progress 199
200 to the landing page. We clearly communicated these restrictions to participants at the 200
201 experiment’s outset. 201

202 **Attention and Engagement Monitoring:** We implemented real-time tracking 202
203 of participant attention using a JavaScript library to detect interface visibility disrup- 203
204 tions.⁴ This system captured actions such as minimizing the browser, switching between 204
205 web pages, or launching other applications. The tracking provided second-by-second 205
206 accountability of participant engagement. 206

207 **Post-Experiment Validation:** After completing the experiment, participants re- 207
208 ported any connectivity disruptions and confirmed their genuine engagement with the 208
209 watching task. We combined these self-reports with our objective tracking measures to 209
210 identify and filter data potentially compromised by distractions or divided attention. 210

211 2.4 Tasks 211

212 2.4.1 Transcription Task 212

213 The transcription task requires transcribing successive CAPTCHAs. CAPTCHAs are 213
214 computer-generated images that contain random letters, numbers and randomly placed 214

⁴See more information about the [Intersection Observer API](#).

215 white spaces.⁵ Each CAPTCHA has a total length of 35 characters, including the white 215
216 spaces. Participants spent on average 41 seconds per CAPTCHA. 216

217 We designed the images to be blurred and distorted by lines and dots added on top 217
218 to increase comprehension difficulty.⁶ The interface always presents these CAPTCHA 218
219 images with a text box and a submit button. When the submit button is clicked, the 219
220 next CAPTCHA image appears and the text input for the previous one is stored. The 220
221 task is identical across conditions, as illustrated in Figure 1. 221

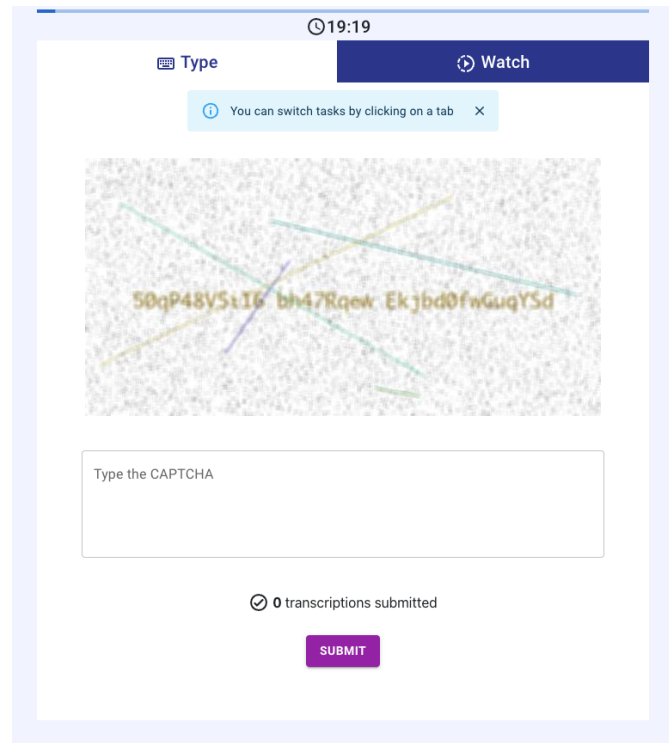
222 We established easy-to-meet quality requirements to ensure participants remained ac- 222
223 tive. These requirements include an overall transcription accuracy of at least 70 percent 223
224 in submitted CAPTCHAs⁷ and a minimum of one CAPTCHA submission per minute 224
225 spent on the task. Participants had to meet these quality requirements to receive com- 225
226 pensation for the transcription task. 226

⁵We added white spaces to improve readability after pretests. The number of white spaces was limited to a maximum of 5 instances per CAPTCHA to control the transcribing time per image.

⁶The [randomization procedure](#) to generate CAPTCHAs.

⁷We use Python's [difflib](#) string-matching algorithm to measure accuracy. It consists of finding the longest common substring and then finding recursively the number of matching characters in the non-matching regions on both sides of the longest common substring. Overall accuracy of 70 percent implies an average of 10.5 ($0.3 * 35$) mistakes across all submitted CAPTCHAs.

Figure 1: Transcription task tab



Note: This figure describes the transcription interface tab. As participants always begin in the transcription tab by default, there is a banner that reminds participants to click the tab buttons to switch tasks. This banner disappeared as soon as participants switched tasks once.

2.4.2 Watching Task

In this task, participants watch successive short videos in a customized media player. Since we are interested in isolating the media player’s default settings, the video content and the order in which videos appear to participants are identical across conditions. We completely disabled the media player controls: Participants cannot skip videos (user scrolling is disabled) or use the media player’s slider to advance or go back in a particular video.

The conditions differ in their autoplay functionality. In the *Autoplay* condition, videos play automatically when the watching tab is visible and pause with a click anywhere on the video. While the watching task is visible, the media player keeps scrolling

to the next videos and continues playing them unless interrupted. In the *Control* condition, participants need to have the watching tab visible and click on each consecutive video to play them. At the end of each video, the scrolling animation brings the next video, but the new video only starts playing when the participant clicks on it. Figure 2 illustrates the video tab in the *Control* condition.

We manually selected 80 videos to isolate the difference in the media player design and help avoid algorithmic biases resulting from the recommender system’s personalized suggestions. We curated videos from YouTube and TikTok, using the tags “funny animal” and “cute animal”. We chose animal videos because of their popularity during pilot studies.⁸ We reviewed the selected videos to ensure that they did not contain any harmful or violent actions toward the animals involved. We kept the first 60 videos shorter in the beginning to maximize the frequency of experiencing the autoplay feature during the watching session, with videos getting longer for the last 20 videos (see Table 2).

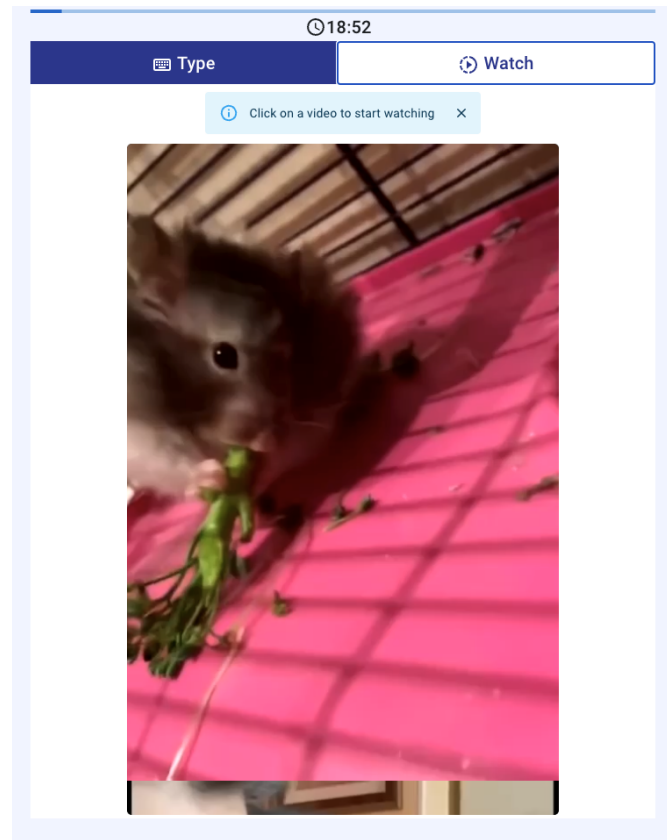
Table 2: Video durations in the watching task

	First 60 videos	Last 20 videos
<i>Duration (seconds)</i>		
Mean (SD)	12.25 (7.04)	25.50 (14.04)
Median [Min, Max]	10 [5, 38]	19.5 [8, 60]

Note: This table presents the duration characteristics of the 80 animal videos used in the experiment. The first 60 videos were kept shorter to maximize participants’ exposure to the autoplay feature during watching sessions, while the last 20 videos were longer.

⁸During the pilots, the experiment was followed by a separate questionnaire asking participants to rate the videos they had seen and suggest what types of videos they would have liked to see more of.

Figure 2: Watching Task tab



Note: This figure describes the “Watch” tab in the *Control* condition. Before the first video is clicked upon to be played, there is a banner that reminds participants that the video won’t play without clicking. This banner was not present in the *Autoplay* condition and videos started playing as soon as participants clicked on the “Watch” button seen up on the right. Notice that the top part of the following video is already visible in the bottom part of the screen.

2.5 Practice Session

The experiment consists of two sessions during which participants interact with the tasks described above. The first session is the practice session on day one. It comes right after the written descriptions of each task and serves to familiarize participants with the tasks and the interface. Participants are assigned to either *Autoplay* or *Control* condition before the practice session, so that they experience the same video interface

on both the Practice and Main Sessions.

During the practice session, each task was available for only 60 seconds and participants were not allowed to switch between tasks. On the first tab, participants were required to retype CAPTCHAs. The 60-second period was enough to submit one CAPTCHA and see the second CAPTCHA image. After 60 seconds, a pop-up appeared on the screen blocking any other interaction and taking participants to the second page.

Participants were required to close this pop-up to advance to the second task. The rationale behind the pop-up was to ensure participants spent equal time on both tasks. Whether or not this pop-up was closed was verified to check participant activity and served as an additional attention check.⁹ Once closed, the watching task began for 60 seconds, enabling participants to watch 4 to 5 short videos depending on whether they were assigned to the *Autoplay* or the *Control* condition. In the *Autoplay* condition, videos played continuously, while in the *Control* condition, videos required a click to start.

We did not remunerate participants for the time they spent here because the purpose of the practice session was to understand the difficulty and pacing of the tasks, as well as navigation within the interface.

2.6 Multiple Price List

In the second block of the experiment, participants made 9 binary decisions (see Figure 3). These decisions were made after the practice session on day one. Participants chose between their preferred media player setting (autoplay off or on) and a bonus payment. We determined bonuses by the per-second earning differences across tasks, multiplied by the expected differences in time spent between the conditions in the first block of the experiment.

Specifically, we expected the average difference between the *Autoplay* and *Control* conditions, in terms of time spent across tasks, to be in the magnitude of 60 seconds (it was 79.5 seconds, from Table 4). This resulted in an average earning difference of

⁹All participants closed this pop-up and switched to the watching task.

284 $60 * (0.15 - 0.1) = 3$ pence. To capture this in the MPL choices offered, we set the 284
285 symmetric bonuses accordingly: £[0.05, 0.1, 0.25, 0.5]. 285

286 Once these decisions were completed, we randomly implemented one of their 9 choices 286
287 and informed participants whether autoplay would be on or off, and of their associated 287
288 bonus earnings. With this knowledge, participants then made their hypothetical time 288
289 allocation for day 2. 289

Figure 3: Multiple Price List page

Autoplay on or off?

In the practice session videos played automatically with **Autoplay**. There is another version **where you need to click on videos to play them**.

You need to make 9 decisions about Autoplay. One of your decisions will be randomly chosen and implemented. **You will receive a bonus payment and watch videos with the chosen Autoplay setting.**

For example, if you choose Autoplay for the first decision and it is implemented, you will have Autoplay and receive an additional £0.5 bonus payment.

 AUTOPLAY +£0.5	 NO AUTOPLAY +£0
 AUTOPLAY +£0.25	 NO AUTOPLAY +£0
 AUTOPLAY +£0.1	 NO AUTOPLAY +£0
 AUTOPLAY +£0.05	 NO AUTOPLAY +£0
 AUTOPLAY +£0	 NO AUTOPLAY +£0
 AUTOPLAY +£0	 NO AUTOPLAY +£0.05
 AUTOPLAY +£0	 NO AUTOPLAY +£0.1
 AUTOPLAY +£0	 NO AUTOPLAY +£0.25
 AUTOPLAY +£0	 NO AUTOPLAY +£0.5

 Please pick your favorite option for each row.

CONTINUE

Note: This figure describes the MPL page where participants made 9 binary choices between their preferred autoplay setting and bonus payments. Each row presents a choice between “Autoplay” (left) and “No Autoplay” (right) with associated bonus amounts ranging from £0.05 to £0.5. The interface required participants to select one option in each row before proceeding. One of these 9 decisions was randomly selected for implementation in the Main Session, determining both the autoplay setting and any bonus payment for day two.

2.7 Time Choice

Following the practice session, participants transitioned to the time choice section. This section required them to specify their preferred time allocation between the two tasks for the following day. Participants were informed the main session would be 20 minutes long and they were free to choose corner solutions, even if that meant focusing solely on one task.

We designed a slider tool with 20-second steps for participants to indicate their time allocation preferences. Initially, the slider appeared grayed out and required a click to activate. This design aimed to ensure participants considered their choices in an active manner. The click-to-activate design had two benefits: It guaranteed participants interacted with the slider and helped participants understand the task-related payoff structure.

To incentivize this decision, we randomly made 5% of the day 1 choices binding in the main session. Binding choices led to payments based on the initial time choice, but only if participants met the quality standards described under Section 2.4.1 during day 2. We excluded data from these participants from the final dataset accordingly. We informed participants whether their allocation was binding with their time choice reminder on day two, right before the start of the main session.

The time choice served as a benchmark for participants, and participants accordingly received a reminder of it on day two. The time choice section offers insights into participants' informed time preferences and allows verifying their commitment to these choices after a 24-hour period.

Figure 4: Time Choice page

Time Choice

Now decide how long you would like to spend **tomorrow** on **Typing** and on **Watching Videos**.

Tomorrow you will have **20 minutes** for both tasks.

1 out of every 20 participant will be **bound** by the choice they made today.

Please **click on the slider** and indicate your **Time Choice**:

i You choose to spend **10 minutes 0 seconds** to **Type**.
 You choose to spend **10 minutes 0 seconds** to **Watch Videos**.
 You would earn **£4.25**.

Tomorrow you will learn whether your **Time Choice** is binding.

CONTINUE

Note: This figure shows the time choice interface with a slider tool allowing participants to allocate time between the two experimental tasks for the following day’s 20-minute main session. The slider initially appears grayed out and requires a click to activate. Real-time feedback shows the selected allocation (here, 10 minutes each) and expected earnings (£4.25). The interface informs participants that 1 out of 20 (5 percent) will be bound by their choice and that they will learn whether their choice is binding on day two.

2.8 Main Session

The main session took place on day two, approximately 24 hours after the time choice. We began by reminding participants of their previous day’s time allocation and informing them whether their time choice was binding. 95% of participants were free to allocate their time on day 2 as they preferred.

The main session lasted 20 minutes. Participants could freely switch between the transcription and watching tasks at any time using the tab interface. Unlike the practice session, there were no time restrictions on individual tasks. Participants had complete autonomy over their time allocation decisions. The interface maintained real-time tracking of time spent on each task, and participants could monitor their time allocation

through the progress bar and a timer.

We implemented the experimental treatment during this session through the media player’s autoplay functionality. Participants assigned to the *Autoplay* condition experienced continuous video playback when the watching tab was visible, while those in the *Control* condition required a click to start each video.

2.9 Hypotheses

User interface design choices are “not separable from other content choices, since their effect depends on where the platform has positioned its content on the underlying content manifold.” (Kleinberg et al., 2022). If content has very low inherent appeal and stickiness, autoplay may have minimal effect on video consumption. An analogy is how introducing breaks would have little impact on engagement for low-stickiness content like documentaries, but dramatically reduce engagement for high-stickiness content like celebrity gossip.¹⁰

Assuming that our selection of videos have sufficient appeal and stickiness, *Autoplay* condition should lead to consumption beyond users’ stated preferences, creating a measurable gap between hypothetical time choice and actual behavior. In this case, we expect autoplay’s behavioral effects to be known by participants, albeit underestimated as to how much they would be affected by these interfaces (Bongard-Blanchy et al., 2021). We thus expect positive demand for a commitment device driven by “sophisticated” participants in line with time preference literature (Ericson & Laibson, 2018).

Hypothesis 1: *Autoplay* condition decreases time allocated to the transcription task while increasing video consumption. We test this using a linear regression:

$$Y_i = \alpha + \gamma \text{Autoplay}_i + \beta \text{TimeChoice}_i + \delta(\text{Autoplay}_i \times \text{TimeChoice}_i) + \epsilon_i \quad (1)$$

where Y_i represents our outcome variables of interest (proportion of time spent on transcription or number of videos watched), Autoplay_i is a binary indicator for the condition, and TimeChoice_i is the standardized proportion of time participants planned to spend

¹⁰See Bao (2025) for recent work on addictive short drama series.

on transcription on day 1. We estimate this model with and without the interaction term to examine whether the autoplay effect varies based on initial planning preferences. We expect $\gamma < 0$ for transcription and $\gamma > 0$ for videos watched.

Hypothesis 2: *Autoplay* condition has an effect on deviations from stated time allocation preferences (actual minus planned time spent on transcription). We test this using a two-sample *t*-test comparing the means between the two conditions.

Hypothesis 3: If participants are aware of their self-control issues, they should exhibit negative Willingness To Pay (WTP) for autoplay, preferring the commitment devices that restrict automatic video transitions. We test this using a logistic regression:

$$Y_{ij} = \alpha + \beta \text{Bonus}_{ij} + \epsilon_{ij} \quad (2)$$

where Y_{ij} is the binary choice of taking autoplay given the bonus j from the MPL for individual i . We calculate the WTP for autoplay as the population-averaged effect of bonus $-\alpha/\beta$, and expect a negative sign indicating preference for commitment against autoplay.

In sum, these hypotheses test whether interface design features can systematically distort optimal time allocation decisions in a setting where participants face monetary incentives favoring the transcription task.¹¹

3 Sample and Procedures

The experiment was run entirely online, using widely available open-source tools.¹² We sampled from the Prolific participant pool, an online recruitment platform commonly used for academic research, and targeted adult British residents. We further imposed the following criteria: Fluency in English, balanced gender representation, and having a stable internet connection (> 30 Mbps on the first and > 25 Mbps on the second day of the experiment). We did not allow mobile devices and restricted access to the experi-

¹¹[Access](#) the preregistered hypotheses. Our preregistration included hypotheses 1 and 3. We have added supporting hypothesis 2 regarding our expectations about the time choice section later.

¹²We used the [MERN stack](#) to develop and host the website in an entirely free way.

ment to Chromium-based browsers to enhance the internal validity of the experiment as previously discussed in Section 2.3. Only participants who succeeded on both attention checks, reported not having engaged in other activities, and did not have connection issues during the experiment were included in the final dataset.

Payment was calculated based on actual time spent on each task during the main session on day 2, with compensation rates of 0.15 pence per second for transcription and 0.1 pence per second for watching. Participants needed to meet the established quality requirements for the transcription task to receive compensation for time spent on that activity. The session concluded automatically after 20 minutes, and participants received their final payment calculation before completing the experiment. Participants received a fixed participation fee of £2.75. Average earning per participant was £4.08, amounting to an average payment of £1.33 (median £1.49) for the main session. Participants who completed all required elements of the experiment received payment within seven days after the second session.¹³

For the first block of the experiment, we preregistered a two-sample parallel design using our prototype data and tested for equality of means across conditions. We found that the required sample size to test Hypothesis 1 with the usual parameter values ($\alpha = .05$, $\beta = .2$) would be 114 per condition, with an effect size of 65 seconds calculated by the difference in the average time spent between conditions ($\mu_{Autoplay} - \mu_{Control}$) and population standard deviation of 165 seconds. Since attrition in longitudinal experiments is commonplace, we invited 301 participants on the first day. On day two, we invited the same 301 participants and received 276 complete responses (91% response rate). No participants failed the attention checks more than twice.¹⁴ We dropped 17 participants who self-reported having connection issues and 24 additional participants who stated having engaged in other activities. We also removed 11 participants who had their time choice enforced randomly. Finally, we removed 40 participants that were detected to spend more than 20 consecutive seconds away from our experimental platform who

¹³Prolific payment policy gives researchers 21 days to transfer funds to participants. The typical delay between study completion and payment is however only 3 to 4 days on average.

¹⁴Minimum requirement by Prolific attention check policy.

397 disproportionately come from the *Autoplay* condition. These 40 participants all reported 397
398 engaging with videos but had their mouse detected outside the experiment window, 398
399 making it impossible to distinguish whether they genuinely watched videos or browsed 399
400 other tabs/windows during the study (see Appendix Table [A.1](#)). The final dataset on 400
401 which we base the analysis consisted of 184 individuals. Table [3](#) shows the demographic 401
402 characteristics of the observed individuals. 402

Table 3: Balance table for the sample

	Autoplay N = 79	Control N = 105	<i>p</i> [A = C]
<i>Gender</i>			
Female	48 (46.15%)	54 (45.00%)	0.510
<i>Age</i>			
Mean (SD)	40.56 (11.66)	40.24 (10.98)	0.725
<i>Employment</i>			
Employed (full time)	62 (59.62%)	67 (55.83%)	0.718
Employed (part time/self)	28 (26.92%)	28 (23.33%)	0.417
Not employed	11 (10.58%)	18 (15.00%)	0.167
<i>Income</i>			
Low (0-15)	28 (26.92%)	34 (28.33%)	0.989
Middle (15-30)	30 (28.85%)	36 (30.00%)	0.505
High (50+)	15 (14.42%)	13 (10.83%)	0.325
<i>Marital Status</i>			
Married/Partnership	56 (53.85%)	60 (50.00%)	0.510
Single	42 (40.38%)	52 (43.33%)	0.534
Previously married	6 (5.77%)	8 (6.67%)	0.927

Note: This table presents demographic characteristics for the sample of 184 participants across conditions. Income categories are presented in thousands of British pounds (£). Employment categories: “Not employed” includes unemployed (looking and not looking) and homemaker. “Previously married” includes divorced, widowed, and separated. *p*-values for binary outcomes are from two-sample tests of proportions; for continuous variables, from two-sample *t*-tests. For employment, part-time and self-employed are together as one category. The sample demonstrates successful randomization with balanced participant characteristics across the conditions.

before and invited 60 participants on day one. On day two, we received 57 complete answers (95% response rate). Once again, no participant failed the attention checks. No participant reported having connection issues. This improvement is likely because we had implemented higher network speed requirements (> 35 Mbps on the first and > 30 Mbps on the second day). We dropped 5 participants who reported having engaged in other activities. As we were interested in the demand for a commitment device, participants did not have their Time Choice enforced in this treatment. We therefore ended up with 52 individuals in the final MPL dataset.

4 Results

4.1 Does Autoplay increase video consumption?

We test our primary hypothesis that autoplay increases video consumption by comparing behavior across conditions. Table 4 presents summary statistics for key outcome variables from the 20-minute main session. Two-sample t -tests reveal no significant differences in any outcome measures between the *Autoplay* and *Control* conditions.

Table 4: Sample statistics for variables of interest

	Autoplay N = 79	Control N = 105	<i>p</i> [A = C]
<i>Time Choice</i>			
Mean (SD)	0.48 (0.31)	0.44 (0.33)	0.410
Median [Min, Max]	0.50 [0.00, 1.00]	0.50 [0.00, 1.00]	
<i>Transcription Proportion</i>			
Mean (SD)	0.65 (0.31)	0.58 (0.33)	0.173
Median [Min, Max]	0.72 [0.00, 1.00]	0.56 [0.00, 1.00]	
<i>Nb. of sessions</i>			
Mean (SD)	5.76 (4.98)	4.89 (5.09)	0.246
Median [Min, Max]	4.00 [1.00, 18.00]	3.00 [1.00, 28.00]	
<i>Session length</i>			
Mean (SD)	456.11 (383.22)	546.35 (411.53)	0.131
Median [Min, Max]	314.75 [68.17, 1212.00]	418.67 [44.93, 1314.00]	
<i>Nb. of CAPTCHA submissions</i>			
Mean (SD)	19.76 (12.11)	17.21 (11.62)	0.150
Median [Min, Max]	19.00 [0.00, 53.00]	15.00 [0.00, 45.00]	
<i>Nb. of videos watched</i>			
Mean (SD)	31.33 (26.34)	32.72 (26.39)	0.723
Median [Min, Max]	25.00 [0.00, 80.00]	34.00 [0.00, 77.00]	
<i>Content rating</i>			
Mean (SD)	6.96 (2.18)	6.62 (2.45)	0.327
Median [Min, Max]	7.00 [2.00, 10.00]	7.00 [0.00, 10.00]	

Note: This table presents key outcome variables from the Main Session. “Time Choice” refers to participants’ stated preferences for transcribing on day 1 (proportion), while “Transcription Proportion” represents the actual proportion of time spent on transcription task during the 20-minute Main Session on day 2. Nb. of sessions is the number of transcription or watching sessions participants completed and Session length represents the mean duration of individual sessions in seconds. Content rating reflects participants’ evaluation of video content on a 0-10 scale. *p*-values are from two-sample *t*-tests comparing experimental conditions.

418 To formally test [Hypothesis 1](#), the effects of the *Autoplay* condition on time spent on 418
419 transcription, we run the regression specified in Equation 1. Our analysis reveals that 419
420 the autoplay manipulation had no statistically significant effect on actual transcription 420
421 behavior. Table 5 presents results for transcription proportion as the dependent variable. 421
422 The coefficient on the autoplay ranges from 0.12 to 0.20 across specifications but remains 422
423 statistically insignificant in all models (p -values > 0.16). Participants' day 1 time alloca- 423
424 tions emerge as a powerful predictor of day 2 behavior: a one standard deviation increase 424
425 in planned transcription time is associated with a 0.67 standard deviation increase in 425
426 actual transcription time ($p < 0.001$). Including day 1 time allocations increases the 426
427 variance explained by the model, rising from 1% to 46%. 427

Table 5: *Autoplay* condition and time spent transcribing

	Transcription proportion (z-score)		
	(1)	(2)	(3)
Autoplay	0.203 (0.147)	0.121 (0.109)	0.119 (0.110)
Time Choice (z-score)		0.673*** (0.057)	0.651*** (0.081)
Autoplay $\times$ Time Choice			0.057 (0.110)
Constant	-0.087 (0.100)	-0.052 (0.076)	-0.053 (0.076)
Obs.	184	184	184
R^2	0.010	0.462	0.463
F -statistic	1.91	77.42	61.52

Note: Robust standard errors in parentheses. All variables are standardized. Time choice represents participants' planned transcription proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Similarly, Table 6 examines the effect on video consumption, a direct test of our primary hypothesis that autoplay would increase video consumption. Across all specifications, the autoplay coefficient is statistically insignificant. This suggests that the autoplay feature did not meaningfully alter participants' video consumption behavior. Again, day 1 plans prove to be the dominant predictor: participants who planned more transcribing watched -0.66 standard deviation fewer videos ($p < 0.001$).

Table 6: *Autoplay* condition and the number of videos watched

	Nb. of Videos Watched (z-score)		
	(1)	(2)	(3)
Autoplay	-0.053 (0.149)	0.029 (0.112)	0.031 (0.112)
Time Choice (z-score)		-0.664^{***} (0.056)	-0.619^{***} (0.079)
Autoplay $\times$ Time Choice			-0.114 (0.105)
Constant	0.023 (0.098)	-0.012 (0.076)	-0.010 (0.076)
Obs.	184	184	184
R^2	0.001	0.440	0.443
F -statistic	0.13	73.06	61.53

Note: Robust standard errors in parentheses. All variables are standardized. Time choice represents participants' planned transcription proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

For completeness, we provide the regression outputs from the unrestricted dataset in Appendix Table A.2 and A.3. These include 40 participants who spent at least 20 consecutive seconds outside either task. Comparing the interaction models (column 3), we observe that the effect of the *Autoplay* condition on transcribing behavior decreases

from 0.119 in the restricted sample to 0.055 in the unrestricted sample. The effect of the *Autoplay* condition on the number of videos watched increases from 0.031 in the restricted sample to 0.132 in the unrestricted sample. However, the null results remain unchanged across both specifications. Because we cannot ascertain the data quality for the unrestricted sample, we base our conclusions on the restricted sample of 184.

We observe two key patterns in the data that help explain the null findings. First, we observe contradicting results: the *Autoplay* condition increases time spent transcribing by 0.12 standard deviations (though statistically insignificant) while simultaneously increasing the number of videos watched by 0.03 standard deviations (also insignificant). If people type more, shouldn't they watch fewer videos? We identified a mechanical confound in our experimental design: the time lost during video transitions. Participants in the *Control* condition, who had to manually click to start each new video, lost an average of 46.52 seconds due to these transition delays whereas those in the *Autoplay* condition lost only 1.08 seconds. This large difference of 45.45 seconds per session is statistically significant ($t(105) = 4.89, p < 0.001$). This difference in time lost not watching videos explains the contradicting results and underscores the importance of looking at the unbiased outcome variable (i.e., number of videos watched) when evaluating the effects of the autoplay feature.

Second, assuming the actual effect size for the number of videos watched lies somewhere between the restricted and unrestricted samples, our power calculations based on pretests were insufficient: The required sample size to provide a two-sided test for [Hypothesis 1b](#) with the usual parameter values ($\alpha = .05, \beta = .2$) would be between 300 to 5000 per treatment, with common standard deviation as the root MSE from the standardized number of videos watched variable in the third regression. Other solutions to this power problem are design-related and involve extending the main-session duration to above 20 minutes, or nudging participants to mix more between tasks to avoid corner solutions (e.g., higher earnings for time allocations between .25 and .75 of either task).

Table 7: Seconds lost by condition

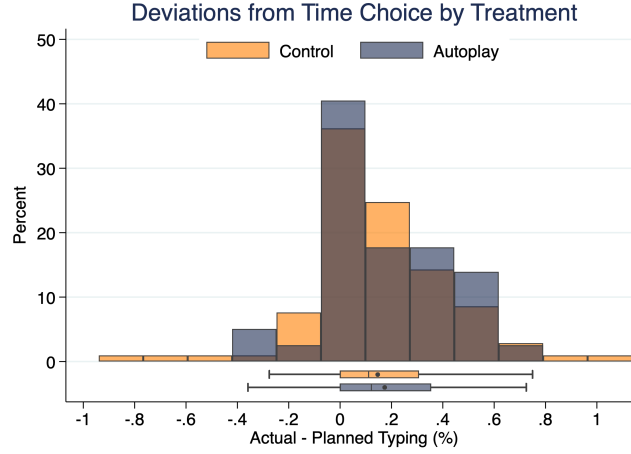
	Control	Autoplay	Difference
Total seconds lost	46.52 (9.27)	1.08 (0.65)	45.45*** (9.29)
Observations	105	79	184
<i>t</i> -statistic			4.89

Note: Standard errors are in parentheses. Time is measured as the average of seconds lost to video transitions per session. The difference was tested using a two-sample *t*-test with unequal variances. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

4.2 Autoplay and deviations from day 1 time allocation

Does the *Autoplay* condition have an effect on deviations from day 1 time choice? To assess [Hypothesis 2](#), we calculated the deviation by subtracting participants' planned transcribing proportion from their actual transcribing proportion (actual minus planned). Figure 5 shows the distribution of deviations from time choice across conditions, which appears right-skewed with positive values indicating that participants transcribed more than their day 1 allocation. Participants in the *Control* condition deviated by an average of 14.6 percentage points, while those in the *Autoplay* condition deviated by 17.3 percentage points. A two-sample *t*-test reveals no statistically significant difference between means ($t = -0.694$, $p = 0.488$). The Kolmogorov-Smirnov test similarly finds no significant difference in distributions across conditions ($D = 0.085$, $p = 0.903$). These results indicate that participants in both conditions spent more time on the transcription task than they initially planned. However, the *Autoplay* condition did not significantly alter the magnitude or pattern of these deviations. Contrary to our hypothesis, we find no evidence that *Autoplay* condition led participants to deviate toward consuming more videos.

Figure 5: Distribution of deviations from day 1 time allocation by condition



Note: This figure shows the distribution of deviations from day 1 time allocation plans, calculated as actual transcribing proportion minus planned transcribing proportion. Positive values indicate participants transcribed more than they planned. The distributions are similar across conditions, and show over-transcribing relative to initial plans.

4.3 Willingness To Pay for autoplay

Contrary to Hypothesis 3, participants exhibited positive WTP for autoplay rather than demand for commitment devices against it. We assessed the demand for commitment by comparing the share of participants choosing to turn autoplay on or off along with the associated bonus payments for each of the nine decisions in the price list. To quantify participants' WTP for autoplay, we estimate the regression specification in Equation 2, where the dependent variable is the binary choice of selecting autoplay (1) or turning it off (0), and the independent variable is the bonus amount in pounds from the MPL.

Results are presented in Table 8. The coefficient on the bonus amount is 27.97 ($p < 0.001$), indicating that participants are sensitive to the financial cost of keeping autoplay on. The WTP for autoplay is the negative ratio of the constant to the bonus coefficient: $-0.6263/27.9729 = -0.0224$ pounds, or 2.24 pence for the 20-minute session. On average, participants value autoplay at 6.72 pence/hour and would be willing to forgo

494 this amount to maintain access to the feature rather than have it turned off. Participants 494
 495 in our setting did not appear to perceive a tradeoff from autoplay in increasing video 495
 496 consumption. 496

Table 8: Willingness to Pay for autoplay

	(1)
Bonus (£)	27.973*** (6.277)
Constant	0.626** (0.199)
Obs.	468
Clusters	52
Pseudo R^2	0.658
Wald $\chi^2(1)$	19.86

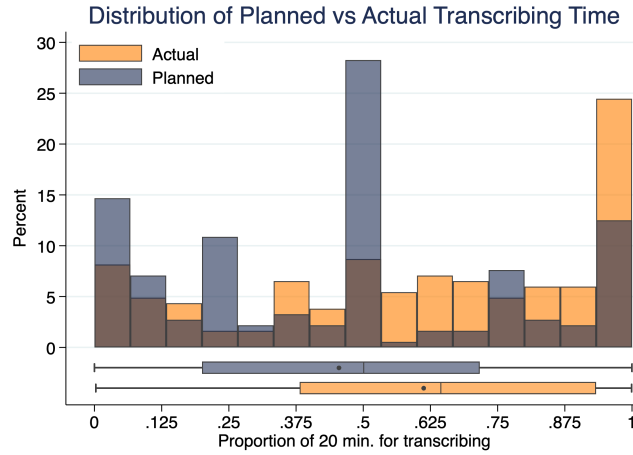
Note: Logistic regression with standard errors clustered at the participant level in parentheses. Dependent variable is binary choice of selecting autoplay (1) or turning it off (0). Bonus amounts range from £−0.5 to £0.5. The pseudo R^2 indicates good fit. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

497 4.4 Understanding the Null Results 497

498 4.4.1 Planned versus Actual Time Allocation 498

499 Participants allocated more time to transcription than initially planned. On average, 499
 500 participants planned to spend 45.5% of their time transcribing but actually spent 61.2%, 500
 501 representing an increase of 15.8 percentage points ($t = 8.24$, $p < 0.001$). The entire 501
 502 distribution shifted rightward (see Figure 6), with median actual transcription time 502
 503 (64.3%) substantially exceeding median planned time (50.0%). 503

Figure 6: Distribution of planned vs actual time allocation



Note: This figure compares the distribution of participants' planned time allocation (Day 1) with their actual time spent transcribing (Day 2). The rightward shift demonstrates participants' systematic tendency to transcribe more than originally planned.

504 The pattern is more pronounced when examining individual transitions. Among 151 504
505 participants who planned to spend less than three-quarters of their time transcribing, 106 505
506 (70%) actually transcribed for three-quarters or more of the session. In contrast, among 506
507 33 participants who planned to spend three-quarters or more of their time transcribing, 507
508 only 4 (12%) reduced below 75%. This asymmetric upward shift contradicts traditional 508
509 models of time inconsistency, which predict succumbing to immediate temptation rather 509
510 than increasing effortful work. 510

Table 9: Transition matrix: Planned vs actual time allocation

Actual (Day 2)	Planned (Day 1)	
	<75% transcription	≥75% transcription
<75% transcription	45 (30%)	29 (88%)
≥75% transcription	106 (70%)	4 (12%)
Total	151	33

Note: This table shows transitions between planned and actual time allocation patterns. The asymmetric pattern demonstrates systematic shift toward transcription, with 106 participants increasing transcription time versus only 4 decreasing it.

511 The deviation toward the more effortful transcription task is rather unexpected, and 511
512 participant perceptions of the decision-making environment may have influenced their 512
513 behavior. This result goes against our hypothesis that participants would deviate toward 513
514 the tempting video content, and points to the need to examine it more closely. 514

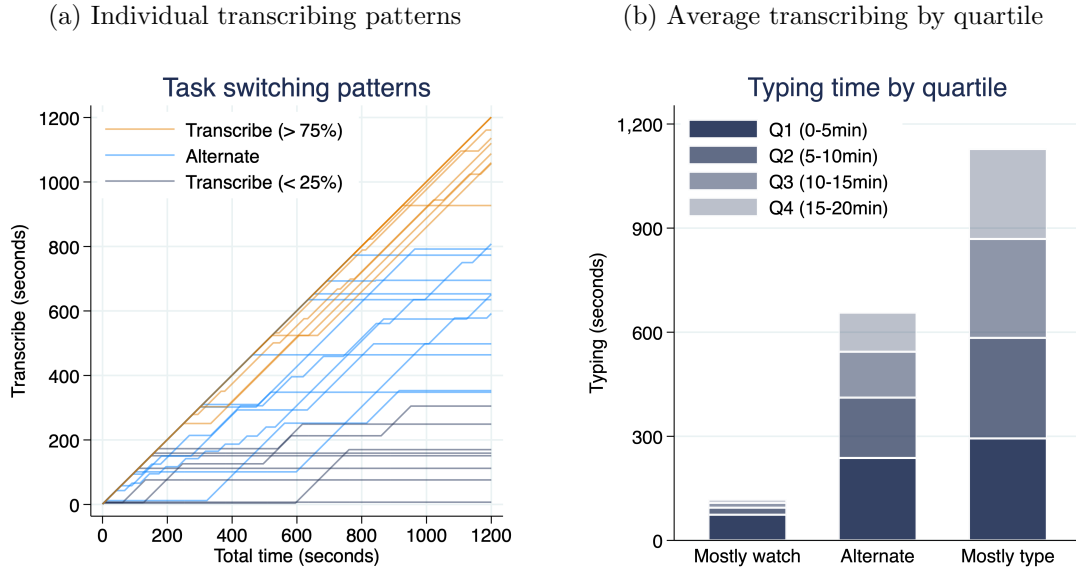
515 4.4.2 Experimenter demand effects 515

516 Why did participants end up transcribing more than their day 1 plans? To understand 516
517 how participants perceived the decision-making environment, we asked how they decided 517
518 to allocate their time during the main session. We analyzed participants’ qualitative 518
519 responses to this open-ended question using dictionary-based concept detection methods 519
520 (Ash & Hansen, 2023). 520

521 We identified words related to negative perceptions of the transcription task (e.g., 521
522 “tiresome”, “frustrating”, “difficult”) and break-taking behavior when switching to watch- 522
523 ing (e.g., “break”, “rest”, “relax”). Figures 7a and 7b show the frequency of these 523
524 concepts in participants’ responses. 524

session. Figure 8a shows substantial heterogeneity in individual transcribing patterns, with transcribing periods interrupted by watching periods for participants who alternate between tasks. Dividing the session into 5-minute quarters reveals that transcribing periods become progressively shorter over time (Figure 8b), suggesting that task switching may be strategic as participants either achieve their day 1 goals or experience fatigue.

Figure 8: Time spent transcribing



Note: Panel (a) shows transcribing time for 30 random participants across the 20-minute session. Panel (b) shows average transcribing time by 5-minute periods, revealing declining transcribing duration over time.

To test whether participants use their day 1 plans as behavioral reference points, we implement a sharp regression discontinuity design using the moment participants reach their planned transcribing duration as the cutoff. We estimate the discontinuous change in the probability of choosing the transcribing task at this threshold.

Results in Table 10 show a significant immediate effect: participants are 27.4 percentage points less likely to continue transcribing in the 30-second window after achieving their plan ($p = 0.007$) and 25.5 percentage points less likely in the 60-second window

($p = 0.003$). This effect dissipates by the 90-second window, becoming statistically insignificant. These findings indicate that participants do treat plan completion as a behavioral milestone, suggesting that experimenter demand effects were not so strong as to completely override their day 1 preferences.

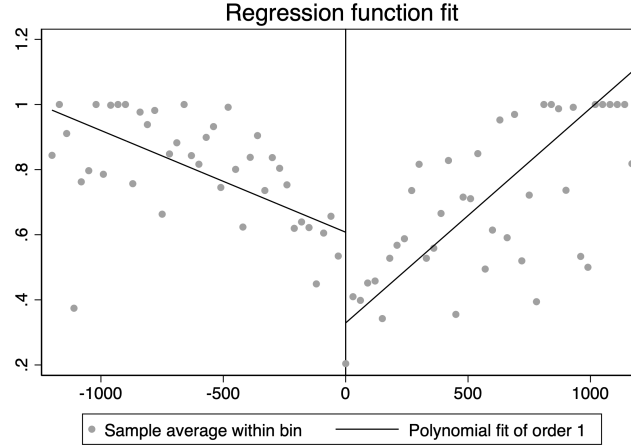
Table 10: Regression Discontinuity estimates on transcribing after achieving day 1 plans

	Time Window		
	30-seconds	60-seconds	90-seconds
Proportion transcribing	-0.274*** (0.101)	-0.255*** (0.087)	-0.075 (0.108)
Obs. (left)	66,786	67,985	67,119
Obs. (right)	80,270	89,006	92,999
Clusters (left)	157	154	155
Clusters (right)	154	157	161
Bandwidth	339.9	368.3	422.3

Note: This table presents robust regression discontinuity estimates of the treatment effect on task choice at the moment participants achieve their day 1 planned transcribing duration. Standard errors clustered by participant in parentheses. The outcome variable is a binary indicator for choosing the transcribing task. Running variables are binned at 30-, 60-, and 90-second intervals around the achievement threshold. All estimates use triangular kernel with MSE-optimal bandwidth selection. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

However, the discontinuity analysis reveals additional complexity. Figure 9 shows that while participants initially reduce transcribing after achieving their day 1 plans, they subsequently increase transcribing effort again. This pattern suggests that participants have indeed found the watching task less engaging than anticipated, leading them to return to transcribing despite having already met their stated preferences.

Figure 9: Linear fit for transcribing rate before and after achieving day 1 plan



Note: This figure plots the average transcribing rate across 30-second bins for participants as they approach their day 1 goal. The negative slope on the left hand side indicates participants working less and less as they approach the cutoff. The drop showcases the discontinuity where achieving the day 1 goal causes in transcribing behavior. The positive slope on the right hand side reveals another trend: As time goes on, participants tend to go back to the transcribing task and decrease their video consumption. This demonstrates participants' tendency to spend more time on the effortful transcribing task than originally planned.

5 Conclusion

This study examined whether autoplay features in isolation can override users' stated preferences for media consumption. In a two-day online experiment with 236 participants, we find no evidence that autoplay increases video consumption when content is held constant. Participants in our setting viewed autoplay positively, exhibiting willingness-to-pay for the convenience it provides. However, systematic deviations from stated preferences suggest experimenter demand effects influenced behavior, with participants transcribing more than planned.

Our null findings reveal a fundamental challenge in studying social media interface features: they are hard to evaluate in isolation from their embedded context. Autoplay's

effectiveness appears to depend on the personalized, engaging content it delivers. Users are not drawn to endless streams of educational documentaries, but to drama, gossip, and emotionally compelling material that algorithms curate personally for them. In our controlled setting with identical and manually curated videos, we may have stripped away the very element that makes autoplay change behavior: its coupling with personalized, addictive content.

These findings have important implications for research on interface design and digital well-being. Laboratory studies like ours that attempt to isolate the behavioral effects of interface features may misrepresent their real-world impacts. The monetary incentives and task framing necessary for internal validity may inadvertently create demand effects that confound the very phenomena researchers seek to study. While autoplay shows no effect in our experimental setting, it seems to have stronger impacts in leisure contexts (Hiniker et al., 2018; Schaffner et al., 2025).

For policymakers considering regulations on autoplay features, our findings suggest that regulating interface features alone may be insufficient if the underlying content and algorithmic curation remain unchanged. Effective digital well-being interventions may need to address the entire ecosystem of personalized, engaging content delivery rather than focusing solely on individual design elements. Future research should prioritize developing experimental methods that better approximate genuine media consumption environments in the field (see e.g., Beknazar-Yuzbashev et al. (2025); Purohit et al. (2023)) while maintaining the control necessary for causal inference.

References

- Allcott, H., Braghieri, L., Eichmeyer, S., & Gentzkow, M. (2020). The welfare effects of social media. *American Economic Review*, 110(3), 629–76.
- Allcott, H., Gentzkow, M., & Song, L. (2022). Digital addiction. *American Economic Review*, 112(7), 2424–2463.

- Arechar, A. A., Gächter, S., & Molleman, L. (2018). Conducting interactive experiments online. *Experimental Economics*, 21(1), 99–131. doi: 10.1007/s10683-017-9527-2
- Ash, E., & Hansen, S. (2023). Text Algorithms in Economics. *Annual Review of Economics*, 15(Volume 15, 2023), 659–688. doi: 10.1146/annurev-economics-082222-074352
- Bao, R. (2025). “just one more clip”: Short videos, big self-control problems.
- Beknazar-Yuzbashev, G., Jiménez-Durán, R., McCrosky, J., & Stalinski, M. (2025). Toxic content and user engagement on social media: Evidence from a field experiment.
- Bhatia, S., Crawford, M. M., McDonald, R. L., Moreno, M. A., & Read, D. (2021). *Inconsistent Planning and the Allocation of Tasks Over Time* (OSF Preprints No. b4mg7). Center for Open Science.
- Bonein, A., & Denant-Boèmont, L. (2015). Self-control, commitment and peer pressure: A laboratory experiment. *Experimental Economics*, 18(4), 543–568. doi: 10.1007/s10683-014-9419-7
- Bongard-Blanchy, K., Rossi, A., Rivas, S., Doublet, S., Koenig, V., & Lenzini, G. (2021). ” i am definitely manipulated, even when i am aware of it. it’s ridiculous!”-dark patterns from the end-user perspective. In *Designing interactive systems conference 2021* (pp. 763–776).
- Braghieri, L., Levy, R., & Makarin, A. (2022). Social media and mental health. *American Economic Review*, 112(11), 3660–3693.
- Bursztyn, L., Handel, B. R., Jimenez, R., & Roth, C. (2023). *When Product Markets Become Collective Traps: The Case of Social Media* (Working Paper No. 31771). National Bureau of Economic Research. doi: 10.3386/w31771
- Chen, Y., Fu, Y., Chen, Z., Radesky, J., & Hiniker, A. (2025). *The Engagement-Prolonging Designs Teens Encounter on Very Large Online Platforms* (No. arXiv:2411.12083). arXiv. doi: 10.48550/arXiv.2411.12083
- Collis, A., & Eggers, F. (2022). Effects of restricting social media usage on wellbeing and performance: A randomized control trial among students. *PloS one*, 17(8),

e0272416.

Ek, C., & Samahita, M. (2023). Too much commitment? an online experiment with tempting youtube content. *Journal of Economic Behavior & Organization*, 208, 21–38.

Ericson, K. M., & Laibson, D. (2018). Intertemporal Choice.

Groshek, J., Krongard, S., & Zhang, Y. (2018). Netflix and Ill? Emotional and Health Implications of Binge Watching Streaming TV. In *Proceedings of the 9th International Conference on Social Media and Society* (pp. 296–300). New York, NY, USA: Association for Computing Machinery. doi: 10.1145/3217804.3217932

Hiniker, A., Heung, S. S., Hong, S., & Kientz, J. A. (2018). Coco’s videos: an empirical investigation of video-player design features and children’s media use. In *Proceedings of the 2018 chi conference on human factors in computing systems* (pp. 1–13).

Hoong, R. (2021). Self control and smartphone use: An experimental study of soft commitment devices. *European Economic Review*, 140, 103924. doi: 10.1016/j.euroecorev.2021.103924

Houser, D., Schunk, D., Winter, J., & Xiao, E. (2018). Temptation and commitment in the laboratory. *Games and Economic Behavior*, 107, 329–344. doi: 10.1016/j.geb.2017.10.025

Kleinberg, J., Mullainathan, S., & Raghavan, M. (2022). The challenge of understanding what users want: Inconsistent preferences and engagement optimization. *arXiv preprint arXiv:2202.11776*.

Kool, W., & Botvinick, M. (2014). A labor/leisure tradeoff in cognitive control. *Journal of experimental psychology. General*, 143(1), 131.

Lukoff, K., Lyngs, U., Zade, H., Liao, J. V., Choi, J., Fan, K., ... Hiniker, A. (2021). How the design of youtube influences user sense of agency. In *Proceedings of the 2021 chi conference on human factors in computing systems* (pp. 1–17).

Lupiáñez-Villanueva, W. F., Boluda, A., Bogliacino, F., & Liva, G. (2022). Behavioural study on unfair commercial practices in the digital environment: Dark patterns

and manipulative personalisation. , 303.

Lyngs, U., Lukoff, K., Slovak, P., Binns, R., Slack, A., Inzlicht, M., ... Shadbolt, N. (2019). Self-Control in Cyberspace: Applying Dual Systems Theory to a Review of Digital Self-Control Tools. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (pp. 1–18). New York, NY, USA: Association for Computing Machinery. doi: 10.1145/3290605.3300361

Lyngs, U., Lukoff, K., Slovak, P., Seymour, W., Webb, H., Jirotko, M., ... Shadbolt, N. (2020). 'I Just Want to Hack Myself to Not Get Distracted': Evaluating Design Interventions for Self-Control on Facebook. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (pp. 1–15). New York, NY, USA: Association for Computing Machinery. doi: 10.1145/3313831.3376672

Mathur, A., Kshirsagar, M., & Mayer, J. (2021). What Makes a Dark Pattern... Dark?: Design Attributes, Normative Considerations, and Measurement Methods. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (pp. 1–18). Yokohama Japan: ACM. doi: 10.1145/3411764.3445610

Nyhan, B., Settle, J., Thorson, E., Wojcieszak, M., Barberá, P., Chen, A. Y., ... others (2023). Like-minded sources on facebook are prevalent but not polarizing. *Nature*, 620(7972), 137–144.

Purohit, A. K., Bergram, K., Barclay, L., Bezençon, V., & Holzer, A. (2023). Starving the newsfeed for social media detox: Effects of strict and self-regulated facebook newsfeed diets. In *Proceedings of the 2023 chi conference on human factors in computing systems* (pp. 1–16).

Schaffner, B., Stefanescu, A., Campili, O., & Chetty, M. (2023). Don't Let Netflix Drive the Bus: User's Sense of Agency Over Time and Content Choice on Netflix. *Proceedings of the ACM on Human-Computer Interaction*, 7(CSCW1), 1–32. doi: 10.1145/3579604

Schaffner, B., Ulloa, Y., Sahni, R., Li, J., Cohen, A. K., Messier, N., ... Chetty, M. (2025). An Experimental Study Of Netflix Use and the Effects of Autoplay on Watching Behaviors. *Proc. ACM Hum.-Comput. Interact.*, 9(2), CSCW030:1–

684 CSCW030:22. doi: 10.1145/3710928 684

685 Silverman, J., Srna, S., & Etkin, J. (2024). *Can Time Limits Increase Time Spent?* 685

686 (SSRN Scholarly Paper No. 4381779). Rochester, NY: Social Science Research 686

687 Network. doi: 10.2139/ssrn.4381779 687

688 Walton-Pattison, E., Dombrowski, S. U., & Preece, J. (2018). ‘Just one more episode’: 688

689 Frequency and theoretical correlates of television binge watching. *Journal of Health* 689

690 *Psychology*, 23(1), 17–24. doi: 10.1177/1359105316643379 690

691 Zizzo, D. J. (2010). Experimenter demand effects in economic experiments. *Experimental* 691

692 *Economics*, 13(1), 75–98. doi: 10.1007/s10683-009-9230-z 692

693 **A Additional Figures and Tables**

693

694 **Additional Figures**

694

Table A.1: Comparison of main sample vs flagged participants

	Main Sample N = 184	Flagged Sample N = 40	<i>p</i> [M = F]
<i>Gender</i>			
Female	89 (48.37%)	13 (32.50%)	0.068
<i>Age</i>			
Mean (SD)	42.29 (11.41)	42.85 (10.75)	0.768
<i>Employment</i>			
Employed (full time)	102 (55.43%)	27 (67.50%)	0.162
Employed (part time/self)	48 (26.09%)	8 (20.00%)	0.420
Not employed	34 (18.48%)	5 (12.50%)	0.366
<i>Income</i>			
Low (0-15)	56 (30.43%)	6 (15.00%)	0.048*
Middle (15-50)	103 (55.98%)	31 (77.50%)	0.012*
High (50+)	25 (13.59%)	3 (7.50%)	0.291
<i>Marital Status</i>			
Married/Partnership	95 (51.63%)	21 (52.50%)	0.921
Single	77 (41.85%)	17 (42.50%)	0.940
Previously married	12 (6.52%)	2 (5.00%)	0.719
<i>Treatment Assignment</i>			
Autoplay condition	79 (42.93%)	25 (62.50%)	0.025*

Note: This table compares demographic characteristics between the main sample (N=184) and participants flagged for potential inattention. Flagged participants reported engaging with videos but had mouse activity detected outside the experiment window, making it impossible to verify genuine engagement. *p*-values for binary outcomes are from two-sample tests of proportions; for continuous variables, from two-sample *t*-tests. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.2: *Autoplay* condition and time spent transcribing

	Transcription proportion (z-score)		
	(1)	(2)	(3)
Autoplay	0.136 (0.133)	0.055 (0.108)	0.055 (0.108)
Time Choice (z-score)		0.595*** (0.061)	0.580*** (0.089)
Autoplay $\times$ Time Choice			0.034 (0.119)
Constant	-0.063 (0.094)	-0.026 (0.078)	-0.027 (0.079)
Obs.	224	224	224
R^2	0.005	0.357	0.357
F -statistic	1.04	50.28	35.00

Note: Robust standard errors in parentheses. All variables are standardized. Time choice represents participants' planned transcription proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table A.3: *Autoplay* condition and number of videos watched

	Nb. of Videos Watched (z-score)		
	(1)	(2)	(3)
Autoplay	0.051 (0.135)	0.131 (0.109)	0.132 (0.109)
Time Choice (z-score)		−0.590*** (0.060)	−0.555*** (0.084)
Autoplay × Time Choice			−0.077 (0.118)
Constant	−0.023 (0.090)	−0.061 (0.075)	−0.059 (0.075)
Obs.	224	224	224
R^2	0.001	0.347	0.348
F -statistic	0.14	48.91	34.21

Note: Robust standard errors in parentheses. All variables are standardized. Time Choice represents participants' planned typing proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.